Coordination Constraints and Information Content in Predicting Lewis Acidity

Simin Huang,* Shenqi Zong, and Han Chen

Tanwei College, Tsinghua University, Beijing 100086, China

*E-mail: hsm24@mails.tsinghua.edu.cn

Abstract

Molecular geometry and electronic descriptors provide complementary routes to predicting the fluoride ion affinity (FIA) of Lewis acids, but their value depends on coordination constraints and the information available about binding. Across 48,986 FIA49k molecules, DimeNet++ achieves a main-test MAE of 11.21 ± 0.27 kJ mol^{-1} over three training seeds, with ensemble analysis showing the largest absolute gain over released connectivity predictions for tridentate ligand environments. P(III) compounds remain difficult across acid-geometry representations, whereas refinement of a PBEh-3c acid–fluoride calculation gives MAE 3.44 kJ mol^{-1} for this class and 3.29 kJ mol^{-1} overall, retaining 3.35 kJ mol^{-1} on held-out ligand-scaffold combinations. Paired molecular structures reveal a second, distinct sensitivity: the loss of accuracy upon changing geometry source is concentrated in acids whose optimized structures disagree, connecting chemical coordination, quantum refinement and structural provenance in the choice of an FIA prediction workflow.

1 Introduction

Fluoride ion affinity (FIA) measures the enthalpic stabilization accompanying fluoride binding to a Lewis acid. It provides a common scale for comparing hard Lewis acidity across the p-block and for identifying promising compounds for anion recognition and bond activation. Quantum-chemical FIA calculations describe both the free acid and its fluoride adduct, including changes in coordination and molecular geometry.[1,2] A predictor built from the free acid must infer this response to binding from the information contained in the unbound molecule.

Several complementary approaches have made this task accessible to machine learning. Sigmund and co-workers introduced FIA49k and FIA-GNN, establishing broad prediction from molecular connectivity across 13 accepting elements.[1] Fenogli and co-workers used chemically informed and substituent-based descriptors to obtain accurate, interpretable predictions for 594 boron Lewis acids distributed over four scaffold families.[3] Their analysis connected substituent position and electronic demand to molecular design. More recently, Mahdian and co-workers quantified the dependence of borane affinity on the probe base, using 3,927 adduct free energies for five bases.[4] Together these studies establish the utility of both broad chemical coverage and focused structure–property analysis. The additional contribution of a resolved three-dimensional acid structure remains an important question, particularly for multidentate ligand frameworks that constrain the accepting center.

Coordination geometry offers a chemical basis for investigating this contribution. Studies of pnictogen pincer compounds show that ligand framework, accepting-center geometry and electronic delocalization jointly affect FIA.[5] Geometric neural networks encode these features through distance filters, equivariant feature messages or directional message passing.[6–8] A second route retains the binding information in a lower-level quantum calculation and learns a correction to a higher-level reference. This Δ-learning principle is well established[9] and has recently been applied to proton affinities using PM7 electronic quantities and cheminformatics descriptors.[10] For Lewis acidity, these two routes ask distinct questions: how accurately binding can be inferred from the acid, and how accurately an existing binding calculation can be refined.

Here we investigate these questions across FIA49k with repeated geometric-model training, electronic-descriptor controls and a separate holdout of ligand-scaffold combinations. Chemical analysis examines the gain from geometry by ligand denticity, the persistent difficulty of P(III), and molecular reorganization in constrained frameworks. Matched-ligand comparisons resolve central-element dependence in the quantum correction. Finally, paired PBEh-3c and GFN2-xTB structures distinguish the chemical difficulty of a molecule from sensitivity to the geometry-generation protocol. These analyses provide a chemical account of where structural learning helps and where explicit acid–adduct information becomes particularly useful.

2 Computational methods

2.1 FIA targets and structural partitions

The target is gas-phase FIA from FIA49k, recorded as fia_gas-DSDBLYP and expressed in kJ mol^{-1}. The reference uses RI-DSD-BLYP-D3(BJ)/def2-QZVPP single-point energies on PBEh-3c structures, with the published thermochemical and anchoring conventions.[1,11] Targets, optimized geometries and released FIA-GNN predictions were obtained from versioned Figshare archives.[11–13] Each geometric predictor receives one PBEh-3c structure of the unbound acid.

The official partition contains 32,333 training and 7,001 validation molecules. Six test groups are evaluated separately: the main FIA44k test (6,732 molecules), CSD compounds (1,200), bimacrocycles (763), four-membered-ring compounds (911), phase-transfer catalysts (15) and additional catalysts (31). All molecules are retained, and predictions are aligned by identifier and reference target. The CSD source contributes compounds to training, validation and test; the two small catalyst groups provide case-based evaluations.

A secondary partition groups the 46,066-molecule training, validation and main-test pool by ligand structure. The accepting center is removed from the grouping key. Cyclic ligands are represented by generic Bemis–Murcko scaffolds and acyclic ligands by generic full graphs. The sorted multiset of ligand representations defines a group, assigned by a fixed-seed hash with nominal training/validation/test probabilities of 0.70/0.15/0.15. The resulting 33,460/6,379/6,227 molecules form 12,448 groups. Complete

scaffold combinations are held out; individual components can recur in different combinations. Models on this partition are independently fitted. An exact complete-ligand-set partition is also documented in the SI.

2.2 Molecular representations and learning

Geometric models use trainable atomic-number embeddings and 5.0 Å radius graphs. The radial-filter model contains four width-128 interaction layers, 32 Gaussian radial functions and a cosine cutoff envelope.[8] Sum pooling is compared with a readout concatenating the accepting-center, mean and sum representations. The accepting center follows the archive's first-atom convention, verified against the recorded element. The EGNN feature-message baseline has seven residual layers of width 128 and fixed input coordinates, following the upstream QM9 implementation.[6] DimeNet++ has four interaction blocks, hidden width 128, interaction width 64, seven spherical and six radial basis functions, and a neighbor cap of 118.[7] Full configurations accompany the frozen checkpoints.

Each neural model is initialized from scratch and trained with AdamW, normalized L1 loss, learning rate 0.001 and weight decay 10-6. Learning rates are reduced after eight epochs without validation improvement. Training ends at 100 epochs or after 25 epochs without improvement, with batch size 64, or 32 for DimeNet++. The checkpoint minimizing validation MAE is selected for each of three random seeds. All target normalization and feature preprocessing use training molecules.

An acid-QM extension combines the geometric representation with PBEh-3c Mulliken and Löwdin center charges, dipole magnitude, electrophilicity, HOMO and LUMO energies, together with four numerical metadata fields. Composition controls use the radial architecture with zero pair distances on complete graphs, with and without these global quantities. Classical controls use histogram-based gradient boosting (HGB) on metadata, 190 varying RDKit 2D descriptors, acid electronic quantities or their combinations.[14,15] Metadata describe the accepting element, oxidation state, period, group, ligand count and denticity. HGB uses learning rate 0.05, at most 31 leaves and L2 regularization 1; stages 100, 400 and 1,000 are compared on validation MAE. Released FIA-GNN predictions provide the published connectivity reference, preserving its original optimization procedure.[13]

2.3 Quantum refinement and chemical diagnostics

Quantum refinement learns the difference between reference and PBEh-3c FIA:

$$\widehat{\mathrm{FIA}}_{\mathrm{DSD}} = \mathrm{FIA}_{\mathrm{PBEh\text{-}3c}} + f_\theta(\mathbf{x}). \quad (1)$$

The HGB correction uses lower-level FIA, metadata, RDKit descriptors and acid electronic quantities. Controls include raw PBEh-3c FIA, a global mean correction, center-class mean corrections and a metadata-based HGB correction, with the same validation stage budget. The lower-level FIA requires calculations of both acid and fluoride adduct. Accordingly, performance is reported separately for acid-based prediction and adduct-assisted refinement.

MAE, RMSE, signed error and pooled R^2 use identical target vectors. Neural results comprise mean MAE and sample SD over seeds, together with separately evaluated unweighted prediction ensembles. Paired bootstrap intervals use 2,000 molecule resamples stratified by central-atom class, conditional on the fitted weights. Exploratory chemical diagnostics use frozen predictions. Ligand denticity follows the released metadata; geometric-displacement bins use training-set percentiles. Central first-shell atoms are assigned by nearest distance within each element count recorded in the binding metadata. For three-coordinate centers, the sum of the three interligand angles describes local geometry. Assigned bond lengths, neighbor indices and all class-level results are exported.

Paired acid/adduct structures are checked for atom-count and element-order correspondence, with the added fluoride at the final position. Heavy-atom pair-distance RMS changes describe binding-induced structural displacement. Center-shell RMS changes use acid heavy atoms within 2.7 Å of the center. These structural diagnostics have units of length; energetic deformation requires a separate energy decomposition. Matched-ligand method comparisons use training records averaged within each canonical ligand-set/center-class cell. Geometry-source sensitivity uses frozen PBEh-trained radial models supplied with GFN2-xTB acid structures, recalculating radius graphs while retaining identities and targets. Each molecule contributes one optimized structure per source.

3 Results and discussion

3.1 Accuracy from acid geometry and electronic information

DimeNet++ gives the lowest mean main-test MAE among the geometric models, 11.21 $\pm$ 0.27 kJ mol^{-1} (Table 1). The EGNN baseline gives 12.08 $\pm$ 0.16; the radial center and sum readouts give 12.22 and 12.39, respectively. The three-seed DimeNet++ ensemble achieves 10.29, compared with 12.11 for released FIA-GNN predictions, a paired reduction of 1.82 kJ mol^{-1} with 95% interval 1.59–2.06. This gain compares the resulting connectivity and optimized-geometry workflows, including their respective training procedures.

Spatial information substantially improves the radial model over its composition control: MAE decreases from 28.70 to 12.22 kJ mol^{-1}. Acid electronic quantities reduce the composition-control error to 20.16, while fusion with the geometric model gives 12.18 $\pm$ 0.04. Geometry and acid electronic descriptors therefore contain overlapping predictive information in this representation (Figure 1a). Connectivity supplies another effective description, as demonstrated by FIA-GNN.

3.2 Multidentate coordination concentrates both the gain and the remaining difficulty

The contribution of geometry depends on the ligand environment (Figure 2a). For monodentate systems, the DimeNet++ ensemble gives MAE 9.07 kJ mol^{-1}, compared with 10.51 for FIA-GNN; the corresponding values are 8.87 and 10.65 for bidentate systems. Tridentate systems give the largest absolute gain: MAE decreases from 16.07 to 13.72, a paired reduction of 2.35 kJ mol^{-1}

Table 1 Main-test MAE in kJ mol^{-1}. Neural values are three-seed mean ± sample SD, except the one-seed composition controls. Ensemble errors use averaged predictions. FIA-GNN uses released predictions.

Model	Inputs	MAE	Ensemble MAE
FIA-GNN	Connectivity	12.11	—
HGB metadata	Metadata	37.57	—
HGB RDKit	Metadata + 2D	18.73	—
HGB acid QM	Metadata + acid QM	23.02	—
HGB 2D + acid QM	Metadata + 2D + acid QM	17.35	—
Composition control	Composition	28.70	28.70
EGNN feature messages	Acid geometry	12.08 ± 0.16	11.04
DimeNet++	Acid geometry	11.21 ± 0.27	10.29
Radial, sum readout	Acid geometry	12.39 ± 0.11	11.22
Radial, center readout	Acid geometry	12.22 ± 0.17	11.15
Composition + acid QM	Composition + QM	20.16	20.16
Geometry + acid QM	Acid geometry + QM	12.18 ± 0.04	11.12

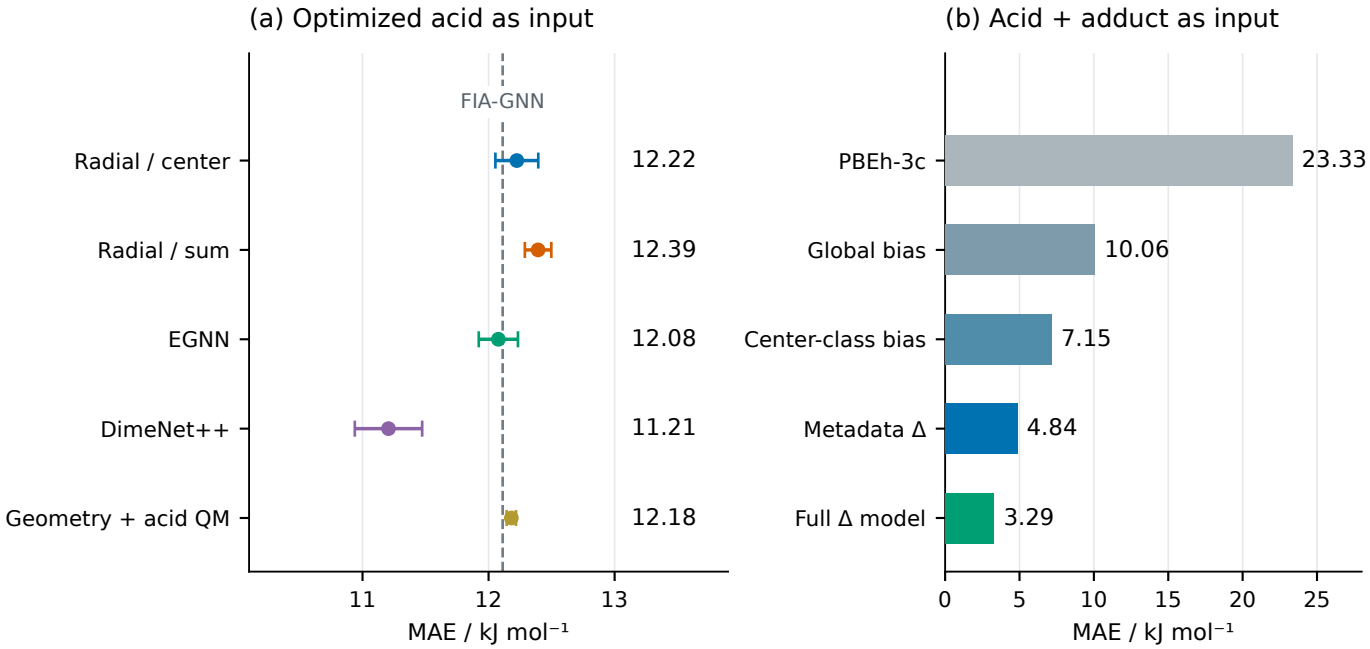


Figure 1 Main-test errors with the required molecular information specified. (a) Optimized-acid models: mean MAE ± sample SD over three seeds; the dashed line marks released FIA-GNN predictions. Values are printed in a separate column. (b) Acid-plus-adduct quantum refinement and training-derived bias controls. Panels represent different input costs.

(95% interval 1.77–2.94; n = 1,875). Resolved geometry is therefore particularly useful in absolute error reduction for the ligand environments that remain hardest to infer from connectivity.

Binding-induced structural displacement provides a complementary view (Figure 2b). The lowest training-defined displacement quartile has DimeNet++ MAE 8.34 kJ mol^{-1}; the upper decile has MAE 12.74. Quantum-refinement errors in the same groups are 2.54 and 4.55, respectively. Structural displacement thus identifies a chemically demanding region for both workflows, while the lower-level binding calculation supplies useful information throughout it. These are exploratory associations across chemically diverse molecules; the descriptor includes peripheral ligand motion as well as coordination-site changes.

3.3 P(III) separates geometric recognition from binding information

P(III) is the most difficult accepting-center class for the DimeNet++ ensemble, with MAE 17.48 kJ mol^{-1} over 219 test molecules. EGNN, radial and acid-QM ensembles give 17.54, 17.71 and 17.76, respectively. The analogous DimeNet++ errors for As(III), Sb(III) and Bi(III) are 12.51, 9.02 and 9.26. The sharp phosphorus dependence persists across distinct acid representations, including explicit angular messages and acid electronic descriptors. In contrast, adduct-assisted refinement gives P(III) MAE 3.44, close to its broad-space error. These P(III) results place a premium on acid–adduct binding information even when an optimized acid structure is available.

Within P(III), geometry is especially useful for multidentate environments. The tridentate subset improves from 20.54 to 15.97 kJ mol^{-1}, with paired reduction 4.57 (95% interval 0.90–8.31; n = 77). The monodentate contrast is less resolved, with its inter-

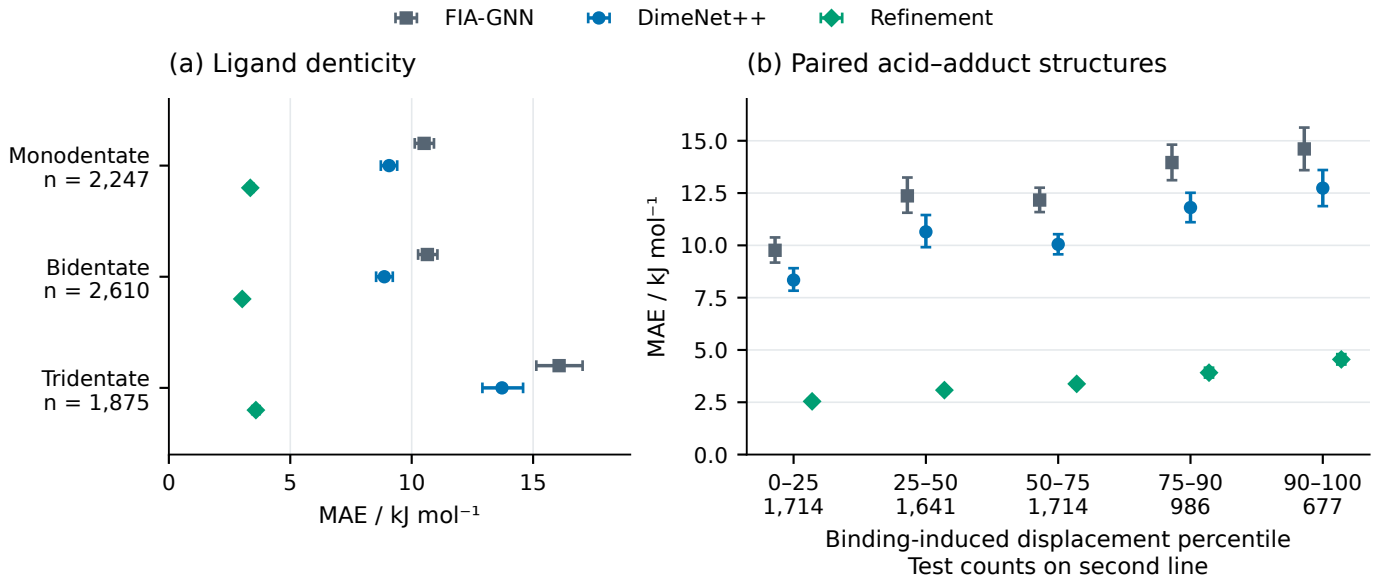


Figure 2 Chemical structure and prediction difficulty on the untouched main test. (a) Ligand-denticity groups. (b) Binding-induced heavy-atom pair-distance RMS change, binned at training-set percentiles 25, 50, 75 and 90; test counts are printed beneath each bin. Symbols denote FIA-GNN (squares), DimeNet++ (circles) and adduct-assisted refinement (diamonds). Intervals are 95% central-class-stratified molecule bootstrap intervals for fixed predictions; adduct coordinates are diagnostic quantities.

val spanning zero (SI). Local angle-sum quartiles likewise retain substantial errors in every bin. This pattern directs attention to the combined coordination and electronic response of a ligand framework upon binding. It complements the geometry–acidity relations established for pnictogen pincers[5] by identifying which information improves prediction across a much broader phosphorus population.

Two held-out molecules illustrate this distinction (Figure 3). P_3_B014413 combines a five-membered P–O-containing ring, an aromatic carbon attachment and chlorinated substituents. Its reference FIA is 252.05 kJ mol^{-1}. FIA-GNN and DimeNet++ give 198.75 and 204.61, whereas quantum refinement gives 250.10. The extended macrocyclic As_3_ZT018778 has reference FIA 406.10; its connectivity, geometric and refined predictions are 267.97, 368.39 and 387.87. The spatial representation recovers much of this macrocycle's affinity, and explicit binding information improves it further. The remaining error makes this compound a useful target for focused quantum-chemical analysis. The examples connect aggregate errors to ligand architecture and remote substitution without assigning an orbital mechanism from prediction residuals.

3.4 Constrained frameworks reorganize at the accepting center

The tridentate boron and phosphorus frameworks in Figure 4 illustrate two geometric responses within tethered ligand systems. The boron framework contains carbon and oxygen attachments, while the phosphorus framework binds through three sulfur atoms. Their molecular drawings expose the ring connectivity; coordination-site depictions show the corresponding optimized free-acid and fluoride-adduct structures under a common projection. Such paired structures make the change in the binding site directly visible.

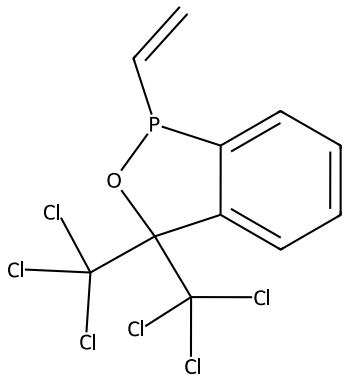
For B_3_ZT009074, the sum of the three ligand–B–ligand angles changes from 359.7ř to 339.5ř, consistent with a nearly planar acid center reorganizing on fluoride coordination. P_3_ZT017132 changes from 316.6ř to 355.7ř, within its sulfur-tethered environment. The reference FIAs are 68.76 and 140.55 kJ mol^{-1}, respectively. These structures illustrate coordination flexibility in constrained acids; both are training-set examples used here for structural interpretation. The thermodynamic contribution of this rearrangement is contained in the relative acid and adduct energies. Its separation into acid deformation and acid–fluoride interaction would require consistent single-point energies at the paired geometries.

The extended arsenic frameworks illustrate the complementary role of the ligand electronic environment (structure atlas in the SI). As_3_ZT018788 and As_3_ZT018778 share three sp2-carbon attachments and the same recorded tridentate macrocycle class, with first-shell angle sums of 279.5ř and 275.2ř. Their reference FIAs nevertheless span 106.54–406.10 kJ mol^{-1}. The dimethylamino-substituted framework has a higher calculated acid LUMO energy and lower electrophilicity than the extensively halogenated framework. This electronic contrast accompanies a large affinity difference despite similar local angle sums, illustrating the information carried by the extended ligand environment. The pair provides a structural comparison; the full denticity and center-class populations establish the predictive trends.

3.5 Central-element corrections persist at matched ligand identity

Quantum refinement resolves a large systematic component of the computational reference difference (Figure 1b). Raw PBEh-3c FIA has main-test MAE 23.33 kJ mol^{-1} and signed error -23.04.

(a) P_3_B014413

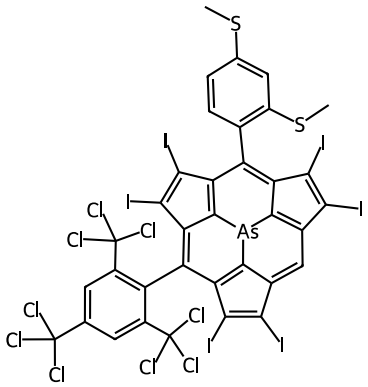


Reference FIA: 252.05 kJ mol⁻¹

FIA-GNN	198.75 (-53.31)
DimeNet++	204.61 (-47.44)
Quantum refinement	250.10 (-1.95)

Prediction (signed error), kJ mol⁻¹

(b) As_3_ZT018778



Reference FIA: 406.10 kJ mol⁻¹

FIA-GNN	267.97 (-138.12)
DimeNet++	368.39 (-37.71)
Quantum refinement	387.87 (-18.23)

Prediction (signed error), kJ mol⁻¹

Figure 3 Structurally distinct held-out Lewis acids: a P(III) heterocycle and an extended As(III) macrocycle. Structures are redrawn from FIA49k SMILES. Both molecules belong to the official main test. Predictions and signed errors are in kJ mol^{-1}; the refinement requires lower-level acid and fluoride-adduct calculations. These illustrations accompany the full-cohort statistics in Figure 2 and the SI.

A global training-set correction lowers MAE to 10.06; center-class corrections give 7.15. The metadata correction reaches 4.84, and the full correction gives 3.29, RMSE 4.49 and R^2 0.9965. Molecular descriptors resolve variation within each center class as well as the mean periodic trends.

Matching exact ligand identity retains a central-element dependence (Figure 5). Across 34 training ligand sets shared by P(III) and As(III), the reference-minus-PBEh-3c correction is 2.66 kJ mol^{-1} larger for As(III), with paired bootstrap interval 0.84–4.62. The corresponding differences are 6.40 kJ mol^{-1} for 98 Ge(II)/Sn(II) pairs and 6.35 for 103 Sn(II)/Pb(II) pairs. Other neighboring-center contrasts are smaller or less resolved. These matched series separate a periodic component of the method correction from changes in ligand population. They provide a chemical reason to condition refinement on both accepting-center identity and molecular environment.

The full correction gives errors of 3.77 kJ mol^{-1} for CSD molecules, 4.03 for bimacrocycles and 3.03 for four-membered rings (Table 2). On the independently fitted ligand-scaffold partition it retains MAE 3.35, compared with 7.25 for center-class bias correction. This transfer across complete scaffold combinations supports a molecular correction that extends beyond the average accepting-center offset. The practical use is refinement of an available PBEh-3c binding calculation to the specified FIA49k reference.

3.6 Geometry provenance introduces a distinct molecular sensitivity

The independently fitted radial-center model gives MAE 12.77 $\pm$ 0.13 kJ mol^{-1} on held-out ligand-scaffold combinations, compared with 19.10 for the matched RDKit control. Acid-QM fusion gives 12.75 $\pm$ 0.03. This evaluates transfer across complete scaffold combinations with shared chemical components; DimeNet++ is evaluated on the official partition. The external-group matrix in the SI additionally shows that bimacrocycles remain challenging for acid-only prediction, with ensemble errors 20.76 for DimeNet++ and 19.89 for acid-QM fusion, compared with 24.08 for FIA-GNN.

A change in structure-generation method creates a separate source of sensitivity. Supplying GFN2-xTB rather than PBEh-3c acid geometries to the frozen radial ensemble increases main-test MAE from 11.15 to 13.02 kJ mol^{-1}. The paired-geometry analysis locates this penalty (Figure 6). In the lower half of the training-defined geometry-disagreement distribution, MAE increases by 1.05 kJ mol^{-1}. Above the training 90th percentile, 0.386 Å pair-distance RMS difference, the increase is 4.91 (95% interval 3.96–5.85; n = 703). PBEh-geometry errors in these groups are similar, 10.70 and 11.02, while xTB-input errors rise from 11.74 to 15.93. The molecular sensitivity to a changed geometry source is therefore distinct from difficulty within the trained structural domain.

This concentration persists after giving each accepting-center class equal weight: the mean penalty is 1.68 kJ mol^{-1} below the training upper-decile threshold and 3.76 above it, across all 21 classes. Paired geometry agreement thus provides a useful crite-

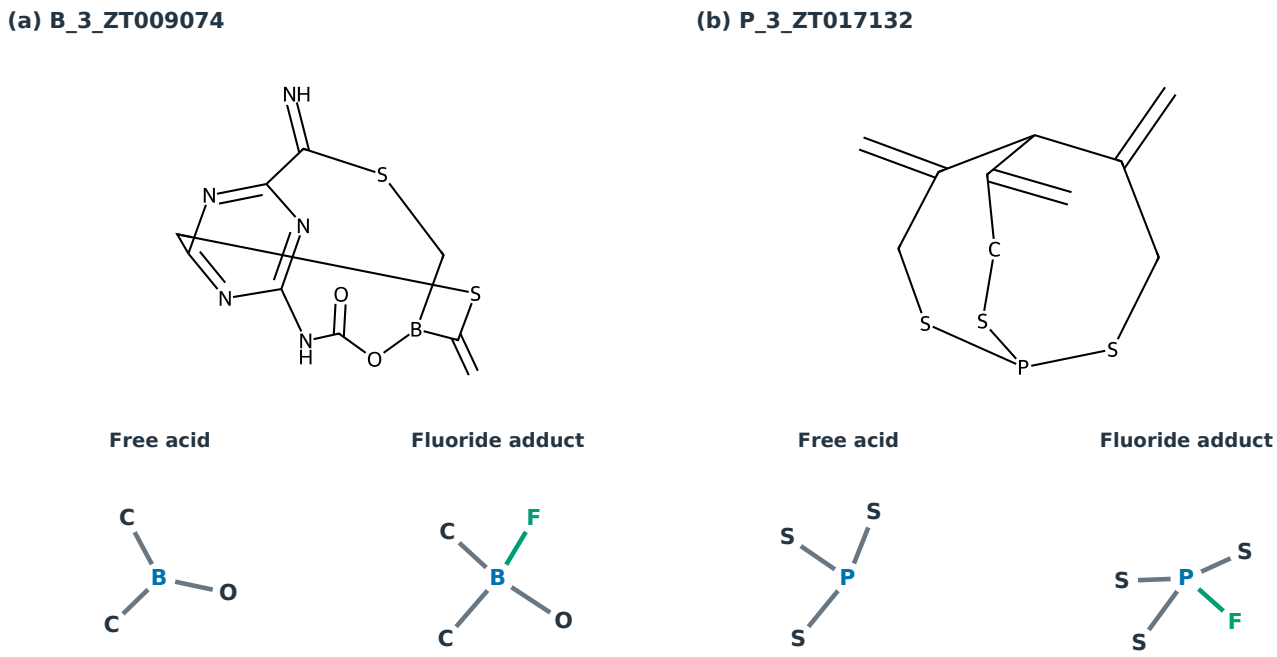


Figure 4 Coordination changes in constrained boron and phosphorus Lewis acids. Complete connectivity drawings are followed by depictions of the accepting center and its three original directly attached atoms in the free acid and fluoride adduct; fluoride is green. Site depictions use released PBEh-3c coordinates. Adducts are rigidly aligned to the three original center–ligand vectors, followed by a shared orthographic projection. Angle sums use the three original ligand atoms; the center-shell distance change is RMS over acid heavy atoms within 2.7 Å. Both molecules are structural illustrations from training.

Table 2 Quantum-refinement MAE in kJ mol^{-1} across the official test groups. All rows use lower-level acid-plus-adduct FIA; correction parameters use training molecules.

Model	Main	CSD	Bimacro	Ring-4	PT	Catalysts
PBEh-3c	23.33	26.73	29.76	26.64	14.19	14.63
Global bias	10.06	10.67	11.31	10.00	9.78	9.74
Center-class bias	7.15	8.41	8.33	6.82	3.23	6.82
Metadata Δ model	4.84	6.94	5.00	3.79	5.37	3.21
Full Δ model	3.29	3.77	4.03	3.03	4.30	2.32

rion for validating a proposed change in the structural workflow. It requires structures from both protocols on the validation compounds. Independent training with the cheaper geometries would address the complementary question of accuracy within that new geometry domain.

3.7 Chemical scope

These results describe three related but distinct information requirements. Connectivity supports rapid broad-space screening. Optimized acid geometry adds the most absolute accuracy in multidentate environments, while the lower-level acid–adduct calculation is particularly effective for chemically difficult P(III) and reorganizing ligand frameworks. Reference-method correction then resolves periodic and ligand-dependent differences between quantum levels. The literature comparison in the SI places these findings alongside broad FIA-GNN prediction, scaffold-focused substituent analysis and proton-affinity Δ-learning.[1,3,10]

The endpoint remains gas-phase fluoride-binding enthalpy at the stated computational reference. Independent FIA benchmarks identify basis-set and anchoring choices as sources of reference uncertainty,[16] and multi-base borane studies demonstrate changes in relative acidity with the interacting base.[4] Generalization to other bases, solution conditions and catalytic outcomes consequently calls for corresponding target data. Within the present endpoint, the chemical analyses identify coordination environments for focused calculations and distinguish structural-domain sensitivity from the information supplied by an explicit binding calculation. Conformer-ensemble benchmarks[17] provide

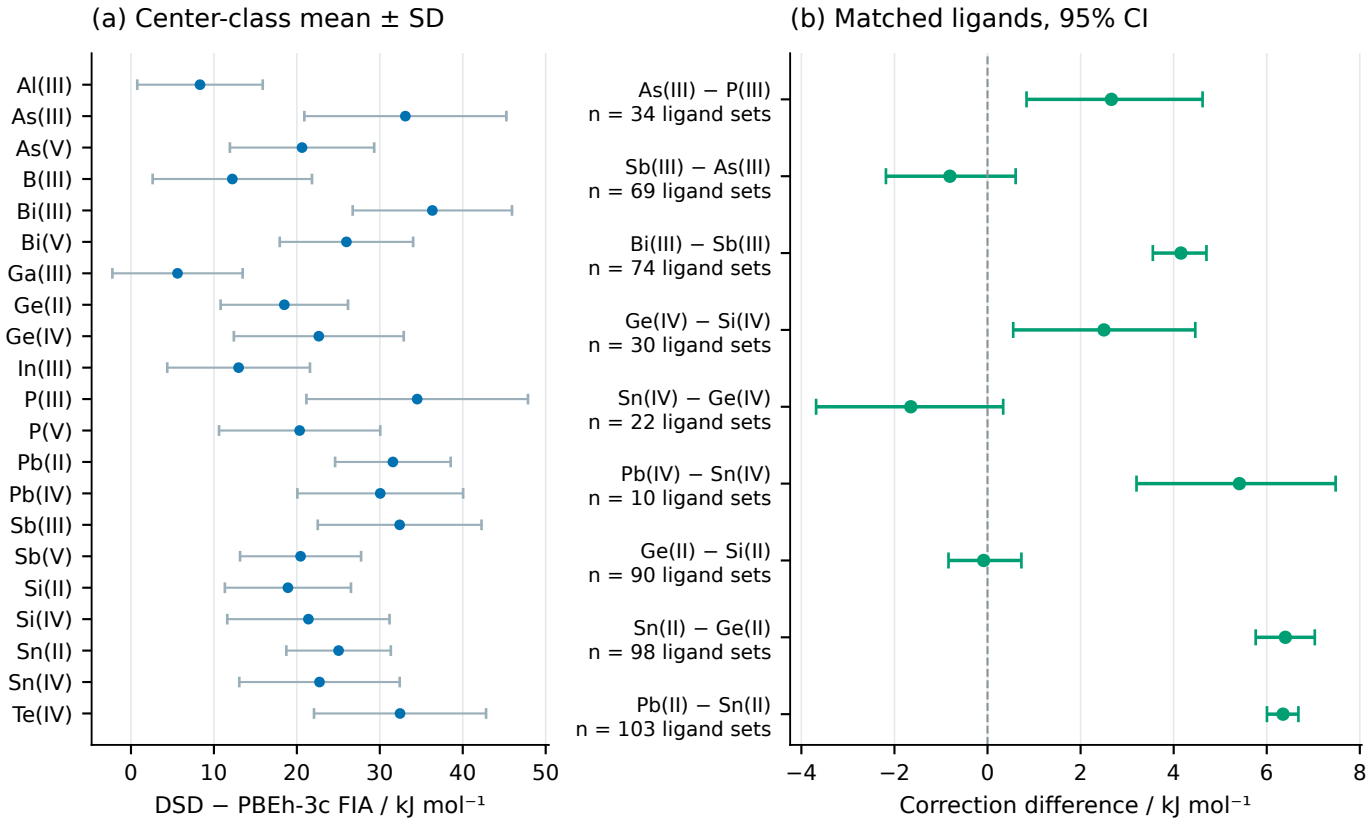


Figure 5 Chemical dependence of reference-minus-PBEh-3c FIA in training. (a) Center-class means with within-class SD. (b) Paired differences between neighboring central elements at identical canonical ligand sets, with 95% ligand-set bootstrap intervals. All nine declared contrasts are shown, including intervals spanning zero.

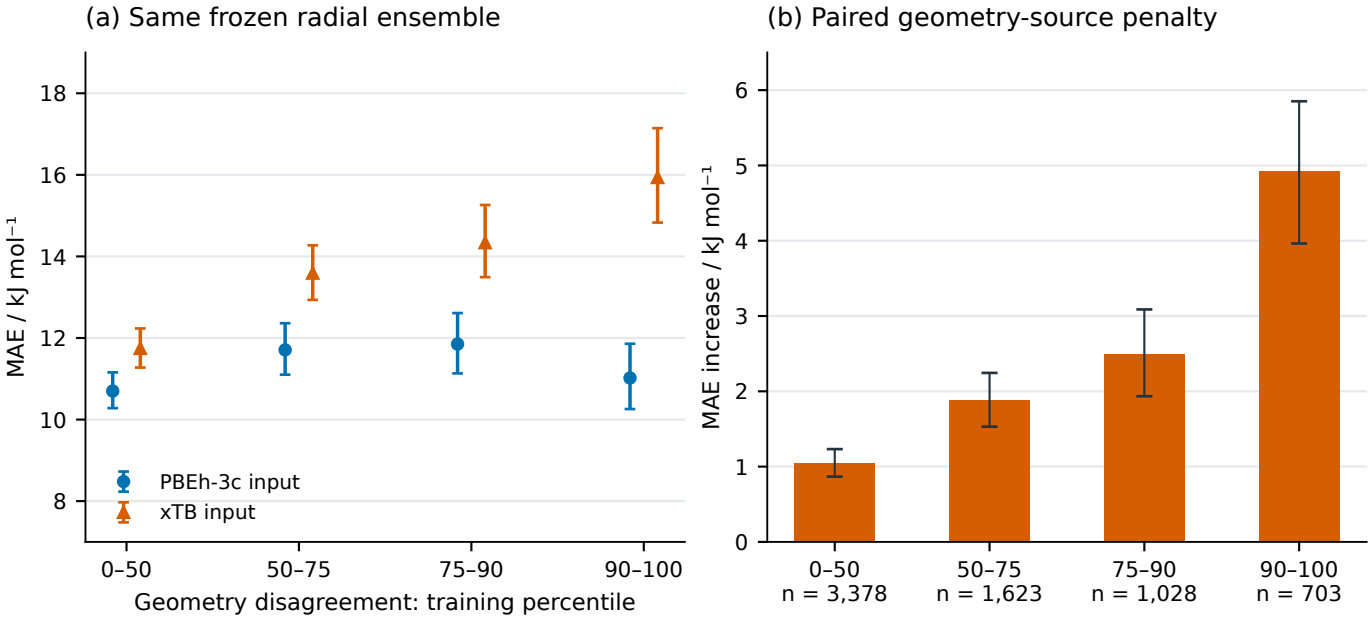


Figure 6 Concentration of geometry-source sensitivity in structurally disagreeing acids. Geometry disagreement is the heavy-atom pair-distance RMS difference between PBEh-3c and GFN2-xTB acid structures; bins use training percentiles. (a) Main-test MAE of the same frozen radial ensemble with the two structural inputs. (b) Paired MAE increase, with test counts and 95% central-class-stratified molecule bootstrap intervals. All weights, identities and reference targets are unchanged.

a complementary route to examine structural sampling within each molecule.

4 Conclusions

Across FIA49k, optimized acid geometry improves prediction most strongly in absolute error for tridentate ligand environments, yet P(III) remains difficult across the tested acid representations. Acid-plus-adduct quantum refinement brings this class close to the broad-space error and transfers across held-out ligand-scaffold combinations. Matched-ligand comparisons identify a periodic component of the reference correction, while paired structures reveal a separate geometry-source penalty concentrated in acids with disagreeing optimized structures. Together, these findings connect model accuracy to coordination constraints, the molecular response to binding and the provenance of structural inputs, providing concrete criteria for selecting and validating Lewis-acidity prediction workflows.

Data availability

FIA49k targets, geometries and released FIA-GNN predictions are available under CC BY 4.0 from Figshare. [11–13] Code, frozen models, complete predictions, split manifests and training configurations are available at `https://github.com/PeanutUp/Lewis-Acidity-Prediction`. Source archives and restored artifacts are verified by checksums.

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